

VerdictEdge

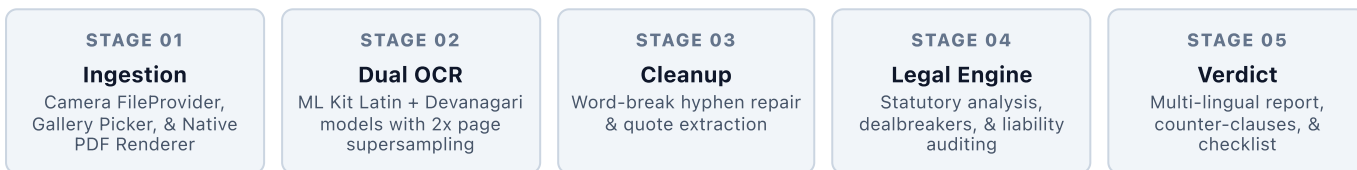
Air-Gapped On-Device Legal Contract Analysis & Statutory Compliance Engine

Package: com.example.clausehawk
Stack: Kotlin 2.0 • Jetpack Compose
Target: Android 10+ (API 26–35)
Security: 100% Offline / Zero Cloud Leaks

Executive Summary: VerdictEdge is an enterprise-grade mobile application designed to evaluate, audit, and de-risk high-stakes legal contracts (NDAs, employment agreements, vendor contracts, service agreements) entirely on-device. By executing optical character recognition (OCR) and clause reasoning locally on the smartphone's hardware, it guarantees zero exposure of confidential legal data to third-party cloud servers.

1 End-to-End Processing Architecture

The application operates on an air-gapped, zero-latency pipeline spanning four foundational stages:



2 Key Features & Capabilities

VISION & FILES

Multi-Format & PDF Ingestion

Ingests high-resolution camera photos, multi-image gallery selections (up to 10 files simultaneously), and native multi-page PDF documents (up to 15 pages per document) rendered via Android PdfRenderer.

INTELLIGENCE

Custom Dealbreaker Enforcement

Allows legal reviewers to specify bespoke redline conditions (e.g., *non-compete*, *60 days notice*, *exclusivity*). Any match immediately elevates the contract risk to High.

STATUTORY LAW

Statutory Voidability Audit

Identifies clauses that may be null and void under statutory jurisprudence, such as post-termination non-competes in restraint of trade under Section 27 of the Indian Contract Act.

NEGOTIATION

Counter-Clause Drafting

Flags predatory or one-sided provisions (broad indemnities, uncapped liabilities) and suggests ready-to-use, balanced replacement clauses for negotiation.

Core Component Architecture

Detailed Module Breakdown

3 Technical Module Breakdown

A. ContractEngine.kt — Legal Evaluation & Risk Analytics

The reasoning kernel executes contract verification, pattern analysis, and risk scoring in Dispatchers.Default:

- **Document Legitimacy Guard (isLegitimateContractText):** Performs rapid structural keyword validation to reject empty, corrupted, or non-legal text files before triggering heavy analytics.
- **OCR Hyphen & Linebreak Normalizer:** Reconstructs words severed across lines by OCR (e.g. indemni- \n fication → indemnification).
- **Sentence-Aware Context Window (extractQuote):** Extracts exact verbatim sentences surrounding detected clauses without truncating mid-sentence.
- **Financial Exposure Auditing:** Flags compounding penalties, daily liquidated damages (e.g. \$50,000/day clauses), and unilateral attorney fee shifters.
- **Discretionary Ambiguity Detection:** Identifies risky subjective clauses containing phrases like "at sole discretion", "reasonable endeavors", or "from time to time".

B. OcrManager.kt — Hybrid Vision & Document Parsing Pipeline

Manages hardware-accelerated local optical extraction without cloud dependencies:

```
// Dual-Script ML Kit Client Initialization
private val recognizer = TextRecognition.getClient(
    DevanagariTextRecognizerOptions.Builder().build()
)

// Multi-URI & PDF Ingestion Flow
suspend fun processMultipleUris(uris: List, onProgress: (Int, Int) -> Unit) {
    - Detects MIME type: application/pdf vs. image/*
    - Uses PdfRenderer to rasterize pages at 2x resolution to ensure small print legibility
    - Runs asynchronous batch extraction with sequential progress notifications
}
```

C. MainActivity.kt — Jetpack Compose State Architecture

The UI is constructed using an unbending 4-state lifecycle machine:

LIFECYCLE STATE	RESPONSIBILITIES	USER EXPERIENCE & CONTROLS
AppState.INIT_LOADING	Cold-start hardware initialization, engine warm-up.	Splash progress bar: "Gemma 2B Initialized • Air-gapped ready".
AppState.INPUT	Document intake, camera capture, file picker, custom rules.	Text area, dealbreaker input, Camera button, Multi-file Gallery button.
AppState.PROCESSING_LOADING	Background OCR and legal contract risk analysis.	Pulsing indicator with clause evaluation progress.
AppState.RESULTS	Report presentation, counter-clauses, pre-signing checklist.	Risk badge, red flags, interactive checklist, history button, and reset.

Risk Models, Localization & Security

Compliance & Future Roadmap

4 4-Tier Risk Classification Matrix

RISK VERDICT	INDICATOR	TRIGGER CONDITIONS	RECOMMENDED ACTION
CRITICAL RISK	CRITICAL RISK	Uncapped indemnity, statutory void provisions (Section 27), custom dealbreakers, liquidated damages.	Do not sign. Propose counter-clauses and negotiate liability caps.
MODERATE RISK	MODERATE RISK	Strict termination notices, consequential exclusions, ambiguous discretionary phrasing, payment deadlines.	Review timelines, verify mutual notice periods, and confirm milestones.
LOW RISK	LOW RISK	Standard mutual NDA, reciprocal termination, balanced intellectual property ownership.	Standard boilerplate. Low legal exposure. Safe to proceed with standard review.
INVALID TEXT	INVALID	Unreadable image, blurry photo, corrupted PDF, or non-legal document.	Re-scan document with better lighting or provide an actual contract.

5 Trilingual Localization & QuickScan Widget

LOCALIZATION Native Trilingual Support All analysis summaries, pre-signing checklists, and risk explanations provide 1-tap dynamic switching between: - English (Standard legal terminology) - Hindi (हिंदी) (Devanagari script support) - Kannada (ಕನ್ನಡ) (Regional language legal synthesis)	SYSTEM WIDGET QuickScan Home Screen Widget Implemented via QuickScanWidget.kt as an Android AppWidgetProvider. Placed directly on the user's home screen for instantaneous 1-tap launching into the document ingestion camera.
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6 Security Model & Future LLM Roadmap

Air-Gapped Security Architecture: Zero Network Transmission: The application does not transmit contract clauses over HTTP/HTTPS. Volatile Context: In-memory session buffers prevent unencrypted disk leaks of proprietary trade secrets. Gemma 2B LiteRT Roadmap: Prepared for offline quantized LLM reasoning using MediaPipe's GenAI SDK (gemma-2b-it-gpu-int4.bin) executing directly on the device NPU/GPU.
